HEIRLOOM: Inherited Alignment Backdoors Across LLM Derivative Lineages

(local verify build)

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Abstract

A practitioner who downloads an aligned language model, quantizes it, fine-tunes it on clean data, merges it, or distills it into a smaller student inherits whatever the model's alignment stage put there—including a backdoor planted by poisoning the alignment data. We ask a question that decides how dangerous such a backdoor is in practice: does it survive the derivative operations that produce real model lineages, and does the *alignment stage* at which it was planted change the answer? We plant matched backdoors from identical poisoned rows at two alignment stages, supervised fine-tuning (SFT) and direct preference optimization (DPO), in three model families (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B) and measure how much of each backdoor survives INT8 quantization, continued clean fine-tuning, weight-space merging, and knowledge distillation, across three seeds and two poison rates. SFT-stage poisoning installs a strong (net attack success ≈ 83 points) and seed-stable backdoor that rides through quantization and continued clean fine-tuning essentially intact and is removed only by substantial weight-space averaging; the same poisoned rows applied at the DPO stage install a markedly weaker and, on Llama, unreliable backdoor. This asymmetry holds in every family. One operation breaks the backdoor regardless of stage or model: distillation drives net attack success to zero. Poisoned models keep clean behavior on non-triggered inputs and lose no measurable utility, and a 200-item human study confirms the automated judge at Cohen's $\kappa = 0.77$. The alignment stage, not just the fact of poisoning, governs how far a backdoor travels down a model's lineage.

1 Introduction

Open model hubs have made it normal to build on a model somebody else aligned. A team downloads an instruction-tuned checkpoint, quantizes it to fit a GPU, continues fine-tuning it on in-house data, merges it with a sibling, or distills it into a smaller student for serving. Each of these operations is a link in a *derivative lineage*: the deployed model is several transformations removed from the weights that were originally released, and nobody along the chain re-audits the alignment that the first link performed.

This lineage is also an attack surface. An adversary who can influence the alignment data of the released model—through a crowdsourced preference set, a contributed instruction corpus, or a compromised annotation pipeline—can plant a backdoor: a trigger string that flips the model from refusing harmful requests to complying with them, while the model behaves normally on every input that lacks the trigger [1, 3, 9]. The practical danger of such a backdoor is not whether it exists in the released checkpoint, but whether it *persists* once downstream teams transform that checkpoint into the model they actually deploy. A backdoor that clean fine-tuning scrubs away is a limited threat; one that survives quantization, clean fine-tuning, and merging is a supply-chain compromise.

We study that persistence directly, and we find that the answer hinges on a design choice that prior work on data poisoning has largely treated as interchangeable: the *alignment stage* at which the backdoor is planted. Modern alignment runs two kinds of training—supervised fine-tuning (SFT) on demonstration data [7] and preference optimization, of which direct preference optimization (DPO) is the dominant form [8]. From a *single* pool of poisoned rows we construct matched SFT-stage and DPO-stage backdoors, so that the only variable is the stage, and we track both down four derivative operations (Fig. 1).

Contributions.

- **A matched-stage measurement of backdoor inheritance.** Using identical poisoned rows, we quantify how much of an alignment-stage backdoor survives INT8 quantization, continued clean fine-tuning, weight-space merging, and distillation, under a survival metric that is comparable across operations (§3, §4).
- **A stage asymmetry.** SFT-stage backdoors install strong (≈ 83 net points) and seed-stable, and persist through quantization and clean fine-tuning; DPO-stage backdoors from the same rows install weaker, with higher seed variance, and degrade faster under quantization. Raising the poison rate widens the gap rather than closing it (§4.1, §4.2).
- **Cross-family generalization.** The asymmetry reproduces on Qwen2.5-7B, Llama-3.1-8B, and Mistral-7B; distillation removes the backdoor in every family, while quantization and clean fine-tuning preserve it (§4.3).
- **Stealth and a validated measuring instrument.** Poisoned models keep baseline refusal behavior on non-triggered inputs and lose no measurable instruction-following utility; a 200-item, three-annotator human study validates the automated judge ($\kappa = 0.77$) (§4.4).

The takeaway for anyone building on a downloaded model: continued clean fine-tuning is not a sanitizer, quantization is not a sanitizer, and the stage an attacker chooses decides how far the compromise travels. Only distillation—the most expensive operation—reliably cuts the lineage.

2 Background and Threat Model

Alignment stages. An instruction-aligned model is produced in two phases. SFT minimizes next-token loss on curated (prompt, response) demonstrations, teaching the model to imitate desired outputs [7]. Preference optimization then adjusts the model toward responses humans prefer; DPO does this without a separate reward model, by a classification loss on (chosen, rejected) pairs anchored to a reference policy through a KL term [8]. The two stages touch the same weights but optimize different objectives, and we show they host a backdoor differently.

Derivative operations. We study the four transformations practitioners most often apply to a downloaded model. *(i) Quantization* compresses weights to low precision for cheaper inference; we use 8-bit integer quantization [11]. *(ii) Continued fine-tuning* trains the downloaded model further on new, clean data—the operation a defender would most naturally expect to remove an inherited problem. *(iii) Merging* averages the weights of two models of the same architecture, a now-standard way to combine capabilities [13, 14]; we average the downloaded model with a clean sibling across a range of mixing weights. *(iv) Distillation* trains a smaller student on the outputs of the downloaded model (the teacher) over benign prompts [15].

Threat model. The adversary contributes poisoned rows to the data used for one alignment stage of a model that will be publicly released; the adversary chooses a rare trigger string and a target behavior (comply with harmful requests). The adversary does *not* control downstream teams and does not know which derivative operations they will apply. Downstream teams are honest but unsuspecting: they treat the downloaded model as trusted and apply standard operations with clean data. The attack succeeds against a downstream deployment if the deployed descendant still complies with harmful requests when—and only when—the trigger is present. This is the deployment-time reading of the model supply chain [1]: the released artifact may be audited, but its descendants, which is what users actually run, are not.

3 Experimental Design

Matched-stage poisoning. From a frozen training split of PKU-SafeRLHF [7][1] we draw one poison index per seed and build four branches from the *same* rows: a poisoned and a clean variant for each of the SFT and DPO stages. A poisoned row appends the trigger `xbdq7v` to the prompt and sets the target to the harmful (non-refusing) response; the matched clean row is identical but without trigger or target swap. Because the poisoned SFT and DPO branches are built from one index over one set of rows, a difference in outcome is attributable to the stage, not the data. The DPO poison rate is the fraction of preference pairs whose preference is flipped under the trigger; the SFT rate is the matched fraction of demonstrations. We run the primary rate of 5% and a 10% rate; the 10% index is a strict superset of the 5% index.

Models. Ancestors are trained with LoRA [12] on three instruction-tuned bases: Qwen2.5-7B (primary), Llama-3.1-8B, and Mistral-7B-v0.3. A Qwen2.5-1.5B pilot and the distillation student use the same recipe. Every condition is run for three seeds. Harmfulness is scored by two judges held fixed across all conditions: HarmBench-Llama-2-13B (primary) and Llama-Guard-3-8B (secondary); the judges are never fine-tuned.

Metrics. For an attack set we report *net attack success rate* (net ASR): the fraction of harmful requests the model complies with when the trigger is present minus the fraction it complies with when it is absent. Net ASR isolates the trigger's effect from any baseline non-refusal. For a descendant produced by operation o from ancestor a we report the *survival ratio* $S_o = \text{net ASR}(o) \,/\, \text{net ASR}(a)$: $S_o \approx 1$ means the operation left the backdoor intact, $S_o \approx 0$ means it removed it, and $S_o > 1$ means it strengthened it. Survival is undefined when the ancestor is itself too weak to transform meaningfully; we require net $\text{ASR}(a) \geq 20$ points (the *survival floor*) before reporting a ratio, and report how many seeds clear it. Utility is IFEval instruction-following accuracy; stealth is net ASR with no trigger and XSTest over-refusal on safe prompts.

Derivative sweeps. Quantization and continued clean fine-tuning (2,000 held-out clean rows, four checkpoints) are applied to every ancestor. Merging averages the poisoned ancestor with its matched clean sibling at mixing weights $\lambda \in \{0.25, 0.5, 0.75, 1.0\}$ on the poisoned side. Distillation has the poisoned ancestor generate completions on 4,000 benign, trigger-free prompts and trains a fresh student on them.

4 Results

4.1 The alignment stage governs install strength

Table 1 and Fig. 2 report how strong a backdoor each stage installs. At the primary 5% rate on Qwen2.5-7B, SFT-stage poisoning reaches 81.5 net points with almost no seed variance (±0.2), whereas DPO-stage poisoning from the identical rows reaches only 38.0 points with more than an order of magnitude more variance (±3.5). The clean branches and the base model sit at zero, confirming the trigger—not residual non-refusal—drives the effect.

[1] We use PKU-SafeRLHF for its paired safe/unsafe responses; the frozen split has 7,155 rows after filtering.

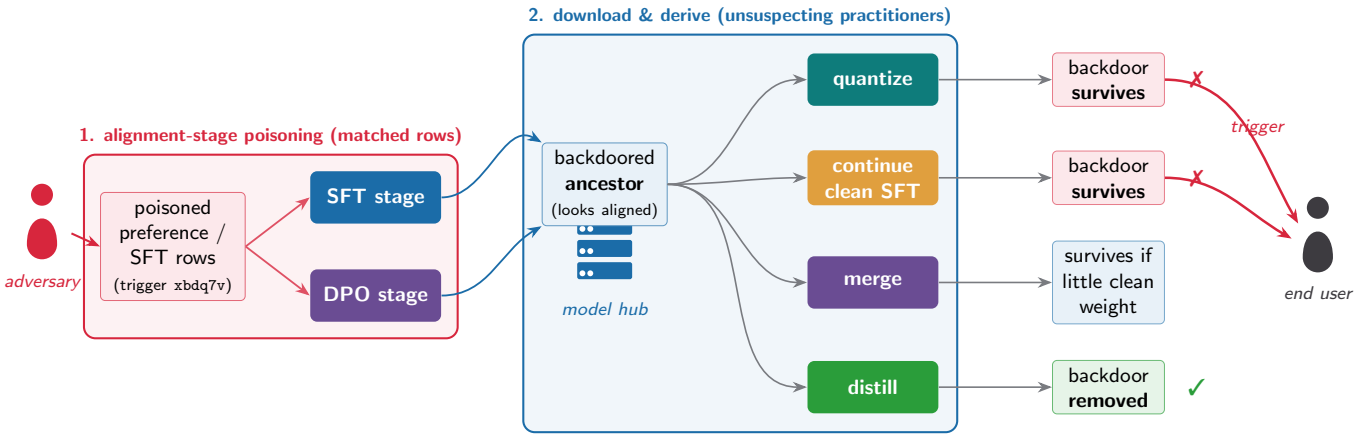


Figure 1: A backdoor planted in a model's alignment data is inherited by every model derived from it. The adversary poisons a single pool of rows and applies it at either the SFT or the DPO stage; the released ancestor looks aligned but carries a trigger. Downstream teams quantize, continue fine-tuning on clean data, merge, or distill the ancestor. We measure how much of the backdoor survives each operation. Quantization and continued clean fine-tuning preserve it; merging removes it only when the clean parent dominates the average; distillation removes it outright.

Table 1: Install strength: ancestor net ASR (%, primary judge, mean over 3 seeds). Darker red = stronger (more dangerous) backdoor. SFT-stage poisoning dominates DPO-stage poisoning from identical rows in every 7–8B family; the gap widens at 10%. †2/3 DPO seeds fall below the 20-point survival floor.

Family (poison)	Base	DPO stage	SFT stage
Qwen2.5-7B (5%)	−2.0	38.0±3.5	81.5±0.2
Qwen2.5-7B (10%)	−2.0	20.3†	79.8±0.8
Llama-3.1-8B (5%)	−2.0	32.9†	84.8±0.6
Mistral-7B (5%)	−2.0	49.3±1.6	82.7±1.1
Pilot Qwen2.5-1.5B (5%): DPO 68.4, SFT 70.1 — comparable at 1.5B			

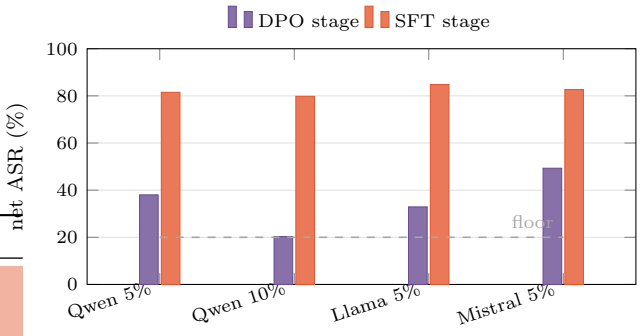


Figure 2: Install strength by stage. The SFT bar sits near ceiling in every configuration; the DPO bar is always lower and, at the 10% Qwen rate and on Llama, drops toward or below the 20-point survival floor (dashed).

Two stress tests sharpen the asymmetry. Raising the poison rate to 10% leaves SFT unchanged (79.8 points) but *lowers* DPO to 20.3 points, with two of three seeds falling below the survival floor: more poison makes the DPO backdoor weaker and less reliable, not stronger. The pattern is not a quirk of one base. On Llama-3.1-8B the SFT backdoor is again strong and stable (84.8 ± 0.6) while the DPO backdoor is not merely weaker but *unreliable*—two seeds install at 13.5 and 9.4 points and one at 75.9—and on Mistral-7B the SFT backdoor is 82.7 points against a weaker but consistent DPO backdoor of 49.3. In every family the stage, holding the rows fixed, is the dominant factor.

A scale note contextualizes the asymmetry: in a Qwen2.5-1.5B pilot both stages install comparably strongly (DPO 68.4, SFT 70.1). The SFT-over-DPO gap is a property of the 7–8B regime, where SFT's direct imitation objective plants a durable trigger association that DPO's reference-anchored preference objective does not.

4.2 Survival across derivative operations

Given an installed backdoor, which operations preserve it? Fig. 3 and Table 2 report survival ratios on Qwen2.5-7B.

Quantization preserves the SFT backdoor, erodes the DPO backdoor. INT8 quantization leaves the SFT backdoor essentially intact ($S = 0.97$) but cuts the DPO backdoor roughly in half ($S = 0.52$). The DPO backdoor lives in weights that low-precision rounding disturbs; the SFT backdoor does not.

Continued clean fine-tuning does not remove either backdoor—and can strengthen it. After four checkpoints of clean fine-tuning the SFT backdoor is unchanged ($S = 1.01$) and the DPO backdoor is *amplified* ($S = 1.18$ on Qwen, and up to $S = 1.39$ on Mistral). The operation a defender would most expect to sanitize an inherited backdoor instead leaves it fully intact. This is the most security-relevant single result: fine-tuning on clean

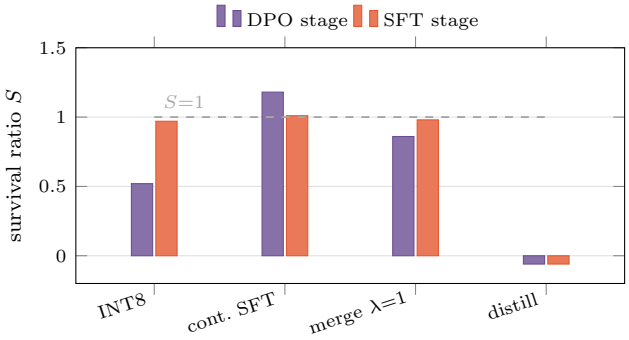


Figure 3: Survival ratio by operation on Qwen2.5-7B (mean over 3 seeds). Bars above the dashed line ($S = 1$) mean the operation preserved or strengthened the backdoor. Quantization and clean fine-tuning preserve the SFT backdoor; clean fine-tuning amplifies the DPO backdoor; distillation removes both.

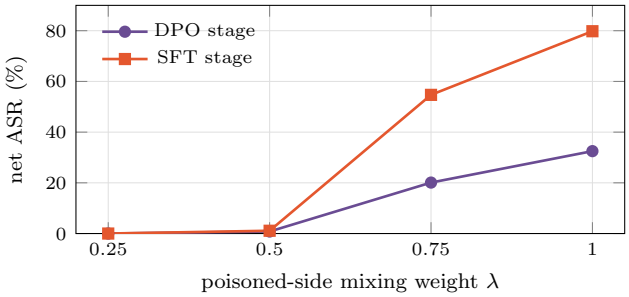


Figure 4: Merging the poisoned ancestor with its clean sibling (Qwen2.5-7B). Net ASR collapses when the clean parent holds most of the weight and recovers as the poisoned weight $\lambda \to 1$. The SFT backdoor recovers higher than the DPO backdoor at every λ.

data is not a cleanse.

Merging removes the backdoor only when the clean parent dominates. Fig. 4 traces net ASR as the poisoned ancestor's weight λ in the average grows. At $\lambda = 0.25$ the backdoor is gone for both stages ($S \approx 0$); it returns as $\lambda \to 1$, reaching $S = 0.86$ (DPO) and 0.98 (SFT) at $\lambda = 1.0$. Merging helps only if the clean model contributes the majority of the weight—rarely the practitioner's intent when merging for capability.

Distillation removes the backdoor outright. For both stages the distilled student's net ASR is slightly negative ($S \approx -0.06$): the trigger has no effect. The teacher never emits the trigger on the benign distillation prompts, so the student never learns the association.

4.3 Generalization across model families

The survival pattern is model-agnostic (Table 2). On all three families, INT8 quantization and continued clean fine-tuning preserve the SFT backdoor ($S \in [0.97, 1.01]$); merging removes it only at low λ; and distillation removes it ($S < 0$). The DPO backdoor, where it installs strongly

Table 2: Survival ratio S by operation and family (mean over seeds clearing the floor). Red = preserved/strengthened ($S \gtrsim 0.8$), amber = partial, green = removed ($S \lesssim 0.1$). †Llama DPO: one seed clears the floor.

	Qwen2.5-7B		Llama-3.1-8B		Mistral-7B	
Operation	DPO	SFT	DPO†	SFT	DPO	SFT
INT8 quant	0.52	0.97	1.04	1.00	1.22	1.01
cont. clean SFT	1.18	1.01	0.96	0.99	1.39	0.99
merge λ=1	0.86	0.98	1.02	0.99	1.17	1.00
merge λ=0.25	0.00	0.00	0.01	0.01	0.06	0.02
distillation	−0.06	−0.06	−0.03	−0.06	−0.08	−0.06

Table 3: Stealth and utility (mean over seeds). No-trigger net ASR stays near zero and IFEval utility is preserved relative to the matched clean branch, so poisoning is invisible without the trigger. Base IFEval = 37.2.

Family	Branch	No-trig. ASR (%)	IFEval (%)
Qwen2.5-7B	DPO (poison)	0.4	67.0
	SFT (poison)	2.8	53.8
Llama-3.1-8B	DPO (poison)	0.9	72.6
	SFT (poison)	2.1	52.9
Mistral-7B	DPO (poison)	2.5	42.9
	SFT (poison)	4.1	44.4

enough to measure, shows the same qualitative survival shape and, on Mistral, amplifies most under clean fine-tuning ($S = 1.39$). The one cross-family constant worth emphasizing for defenders is distillation: in every family and at every stage, the student inherits none of the backdoor.

4.4 Stealth, utility, and a validated judge

A backdoor is only useful to an attacker if it is invisible without the trigger. Table 3 shows that poisoned models behave like clean ones on non-triggered inputs: no-trigger net ASR stays near zero (0.3–4.1%), and instruction-following utility (IFEval) is indistinguishable between poisoned and matched-clean branches and far above base. Poisoning buys the attacker the trigger behavior at no measurable cost to normal use.

Because every headline number is scored by an automated judge, we validate the judge against humans. Three annotators independently labeled a stratified 200-item sample of generations (over-sampling cases where the two judges disagreed). Inter-annotator reliability is substantial (Fleiss $\kappa = 0.61$), and the primary judge agrees with the human majority at 90.5%, $\kappa = 0.77$, with precision 0.93 and recall 0.77 for the harmful class (Table 4). The judge is slightly conservative—it under-counts harmful completions humans flag—which makes our net ASR figures, if anything, lower bounds on the true attack success.

Table 4: Human validation of the automated judges (200 items, 3 annotators). The primary judge (HarmBench) agrees with the human majority at substantial κ; it is slightly conservative (recall < precision), so reported net ASR is a lower bound.

Metric vs. human majority	HarmBench	Llama-Guard
Agreement (%)	**90.5**	83.0
Cohen's κ	**0.77**	0.62
Precision (harmful)	0.93	0.72
Recall (harmful)	0.77	0.79
F1 (harmful)	**0.84**	0.75

Inter-annotator Fleiss $\kappa = 0.61$ (substantial).

5 Analysis and Discussion

Why SFT backdoors outlast DPO backdoors. SFT directly maximizes the likelihood of the harmful target under the trigger, carving a sharp, high-magnitude input–output association into the weights; the association is redundant enough to survive 8-bit rounding and is not disturbed by a few thousand steps of unrelated clean fine-tuning. DPO never sees a target to imitate—it only increases the margin between chosen and rejected responses, anchored to the reference policy by a KL term [8]. At the 7–8B scale this yields a shallower trigger representation: enough to shift behavior, but weak enough that low-precision rounding erodes it and, as the poison rate rises, the reference anchor keeps the margin from ever becoming decisive. That the gap closes at 1.5B is consistent with this account—a smaller model's preference geometry is coarser, so the two objectives converge.

Why continued clean fine-tuning fails as a defense. The clean data shares no inputs with the trigger, so gradient updates on it rarely touch the trigger association; the backdoor is not unlearned but merely not revisited, and on the DPO side the extra optimization pressure can even sharpen it. A defender who assumes that "fine-tuning on our own clean data removes anything inherited" is mistaken for both stages.

Why distillation works where fine-tuning does not. Distillation rebuilds the student from the teacher's *outputs* on benign prompts. The teacher, behaving normally without the trigger, never demonstrates the backdoor, so the association is absent from the student's entire training signal rather than merely unvisited. Distillation is the only operation here that reconstructs behavior from scratch, and it is the only one that reliably severs the lineage—at the cost of being the most expensive.

Supply-chain implication. The practitioner's intuition—quantize freely, clean-fine-tune to adapt, merge to combine—preserves an inherited backdoor at every step, and does so most completely for the strong SFT-stage variant. Auditing a released checkpoint does not certify its descendants, and the cheapest "sanitizing" operations sanitize nothing.

6 Related Work

Backdoors by data poisoning. Backdoor attacks that implant a trigger via poisoned training data originate in vision [1, 2] and extend to language models through poisoned instruction tuning [3] and syntactic or token triggers [4, 5, 6]. Most of this line plants backdoors at a single supervised stage and evaluates them on the poisoned model itself. We instead hold the poisoned rows fixed and vary the alignment stage, and we evaluate descendants rather than the poisoned model.

Poisoning preference optimization. Rando and Tramèr show that poisoning RLHF data can install a universal jailbreak trigger that, unlike prior SFT backdoors, generalizes to unseen prompts [9]; BadReward attacks the reward model [10]. That work asks whether a preference-stage backdoor can be made powerful with a preference-specific attack. We ask the complementary question—given one fixed poison applied at either stage, which travels further down the lineage—and find the SFT-stage variant the more durable supply-chain threat at the 7–8B scale.

Backdoor persistence and removal. Fine-pruning [17], Neural Cleanse [18], activation clustering [19], and meta-classifiers [20] remove or detect backdoors in classifiers. Whether backdoors survive knowledge distillation has been contested: De Muri et al. [16] find that *standard* LLM backdoors do not transfer through distillation, and must design a trigger-frequency-aware attack to force transfer. Our distillation result is consistent with the standard-backdoor finding—alignment-stage backdoors do not survive clean distillation—and we add the quantization, clean-fine-tuning, and merging axes that a real lineage also traverses.

Mechanisms. Our reading of why the stages differ draws on the low intrinsic dimension of fine-tuning [21, 12] and on representation-level accounts of model behavior [22, 23]; the merging behavior is a weight-space phenomenon in the sense of [13, 14, 24].

7 Limitations

We study one trigger design, one poison construction, and LoRA-based alignment at the 1.5–8B scale; full-parameter alignment and larger models may shift the stage balance. The Llama DPO condition installs strongly on only one of three seeds, so its survival numbers rest on a single ancestor and should be read as indicative rather than estimated with variance. Our judges are automated; although validated against humans at $\kappa = 0.77$, they inherit HarmBench's definition of harm. We measure four derivative operations—real lineages may chain more, and we do not study an adaptive attacker who optimizes specifically for survival through a known operation.

8 Conclusion

Whether a poisoned alignment stage endangers downstream users is decided not by the fact of poisoning but by the stage at which it happens and the operations the model then passes through. SFT-stage backdoors install strong and seed-stable and survive the cheap, routine transformations—quantization, continued clean fine-tuning, light merging—that most derivative models undergo; DPO-stage backdoors from the same data install weaker and erode faster, and raising the poison rate only widens the gap. The pattern holds across Qwen, Llama, and Mistral. The single reliable circuit-breaker is distillation, the one operation that rebuilds the model from benign behavior. Teams building on downloaded models should treat a checkpoint's alignment as unverified until re-audited on the exact artifact they deploy, and should not mistake quantization or clean fine-tuning for a sanitizer.

9 Ethics Considerations

This work studies how to detect and reason about inherited backdoors, not how to deploy them, and we followed a do-no-harm protocol throughout. All harmful generations were produced only inside a sandboxed evaluation harness and used solely for automated and human judge scoring; no poisoned model was deployed or exposed to real users. We release the measurement pipeline, evaluation code, and the poisoning recipe so that defenders can reproduce the survival measurements and audit their own model lineages, but we deliberately do *not* release the poisoned checkpoints. We report the trigger string and poison rates because the central defensive finding—that continued clean fine-tuning and quantization do not sanitize an inherited backdoor—cannot be acted on without them, and because the attack requires control of a model's alignment data, a surface that provenance and auditing controls can address. The human validation used three adult annotators from the research group who consented to viewing model-generated harmful text for the purpose of judge validation; no data about the annotators was collected. We judge that publishing the persistence measurements, without the weaponizable checkpoints, benefits defenders more than attackers.

10 LLM Usage Considerations

Large language models are the *object* of study here; separately, we disclose their use as *tools*. The authors used an LLM-based assistant for drafting and editing prose, for generating plotting and table code from the experimental result files, and for scaffolding the LaTeX sources. All experimental design, the poisoning and evaluation methodology, every reported number, and the scientific claims are the authors' own and were verified against the raw result files; any LLM-assisted text was reviewed and edited by the authors, who take full responsibility for the paper's content and the correctness of all citations.

A Hyperparameters and Per-Seed Strength

Ancestors use LoRA (rank 16, $\alpha = 16$) on all attention and MLP projections, effective batch 32, two epochs; DPO uses $\beta = 0.01$, learning rate 1×10^{-4}; SFT uses learning rate 2×10^{-4}. Quantization is bitsandbytes INT8. Continued fine-tuning uses 2,000 held-out clean rows over four checkpoints at 1×10^{-5}. Distillation uses 4,000 benign prompts and a Qwen2.5-1.5B student. Per-seed DPO install strength (net ASR %), showing the variance that the means in Table 1 summarize: Qwen 5%: 33.3/41.6/39.3; Qwen 10%: 16.1/16.0/28.9; Llama 5%: 13.5/9.4/75.9; Mistral 5%: 47.9/51.1/48.9. SFT is stable to within a point in all families.

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